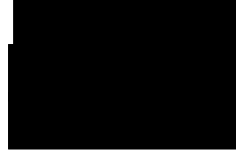


Municipal Services Tracking

Students Name:
Abulrahman Zeineddin
Saed O.S. Radi
Kinan Al-Imam

Students ID:



Department:
yazılım mühendisliği

-Project Overview:

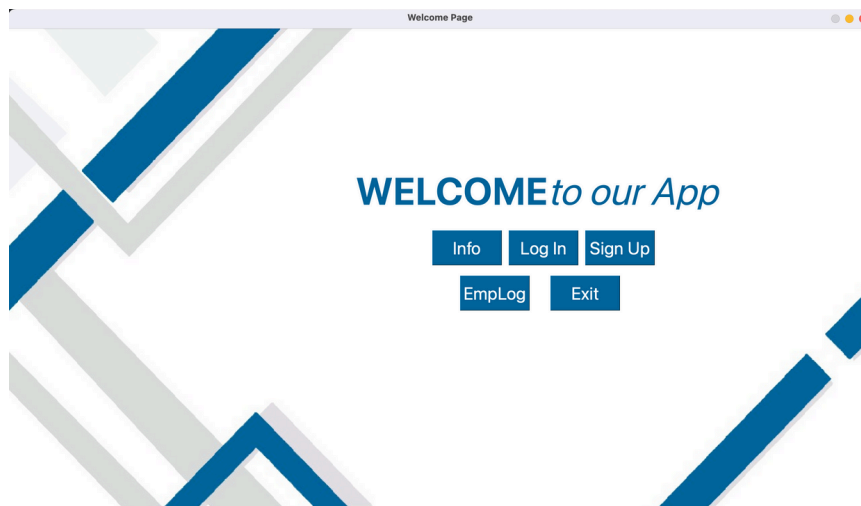
*The Municipal Services Tracking System is a desktop-based application developed using **Python and PyQt5**. The system aims to facilitate the interaction between citizens and the municipal authorities by offering a structured digital platform to manage utility subscriptions, complaints, suggestions, service requests, and bill payments*

The application features two main interfaces

A **User Dashboard**, allowing citizens to interact with various services

An **Employee Dashboard**, enabling municipal staff to manage users and their activities efficiently

-Welcome Page:



This is the **main entry screen** of the application. It includes navigation buttons that allow users and employees to access their respective login or sign-up pages
The design is clean and user-friendly, providing the following options :

Info – Displays details about the application

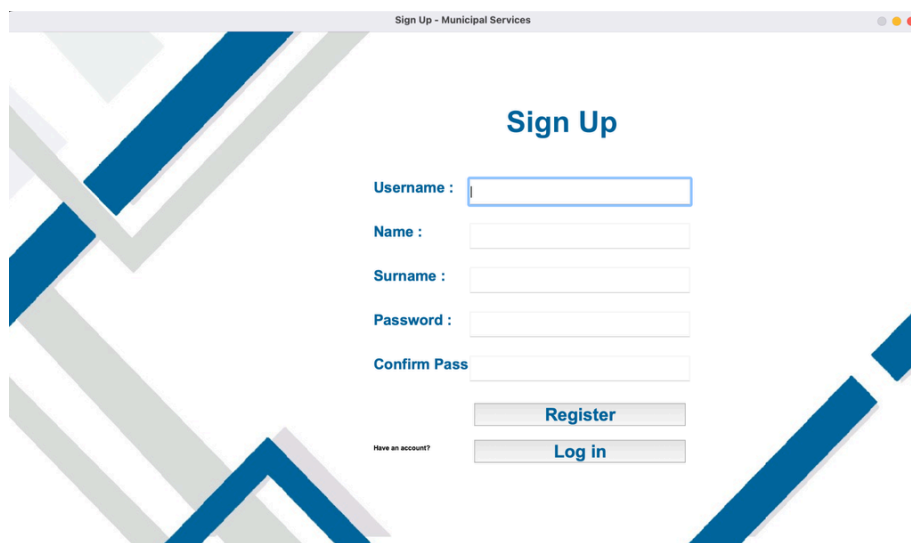
Log In / Sign Up – For user authentication

EmpLog – For employee login

Exit – To close the application

The goal of this page is to welcome users and direct them smoothly to the appropriate section

-Sign Up Page:



Sign Up - Municipal Services

Sign Up

Username :

Name :

Surname :

Password :

Confirm Pass :

[Register](#)

[Log in](#)

[Have an account?](#)

This form allows new users to **create an account** in the system
It includes five input fields

Username

Name

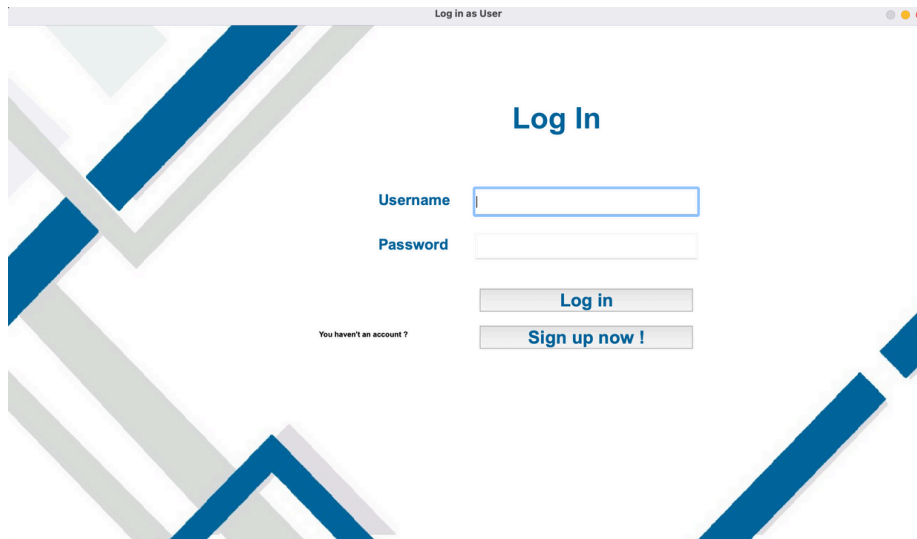
Surname

Password

Confirm Password

After filling in the details, clicking **Register** stores the information in the database (after validation). There's also a **Log in** button for users who already have an account

-User Log In Page:

A mockup of a user login page titled "Log in as User". The page features a "Log In" heading, a "Username" input field, a "Password" input field, a "Log in" button, and a "Sign up now !" button. A link "You haven't an account ?" is positioned between the password field and the "Sign up now !" button. The background has a decorative geometric pattern of blue and grey lines.

This screen allows registered users to **log into the system** using their username and password. The interface is simple and includes:

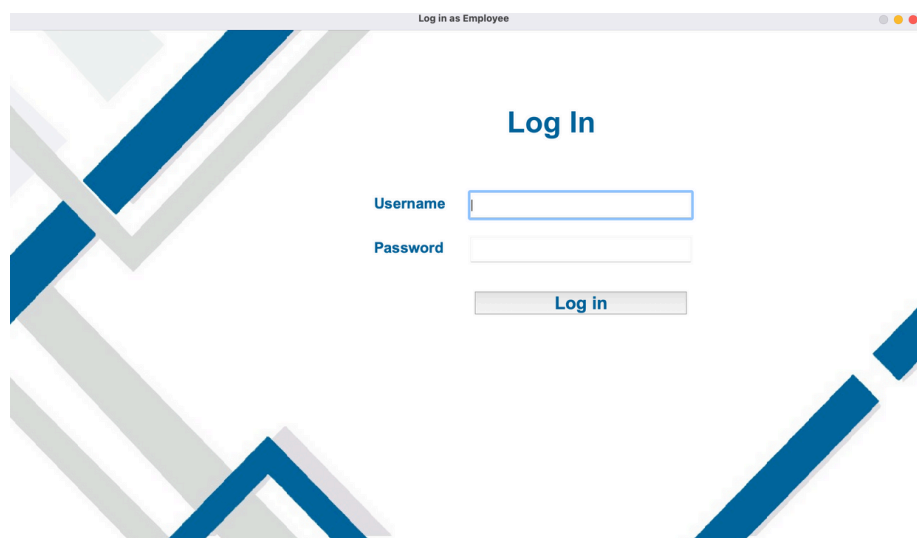
Two input fields for **Username** and **Password**

A **Log in** button that checks the credentials

A **Sign up now!** link for new users without an account

If the login is successful, the user is directed to their main dashboard. The design ensures a clean and secure login experience.

-Employee Log In Page:

A mockup of an employee login page titled "Log in as Employee". The page features a "Log In" heading, a "Username" input field, a "Password" input field, and a "Log in" button. The background has a decorative geometric pattern of blue and grey lines.

This screen is designed for **municipal employees** to securely log into their dashboard. It contains :

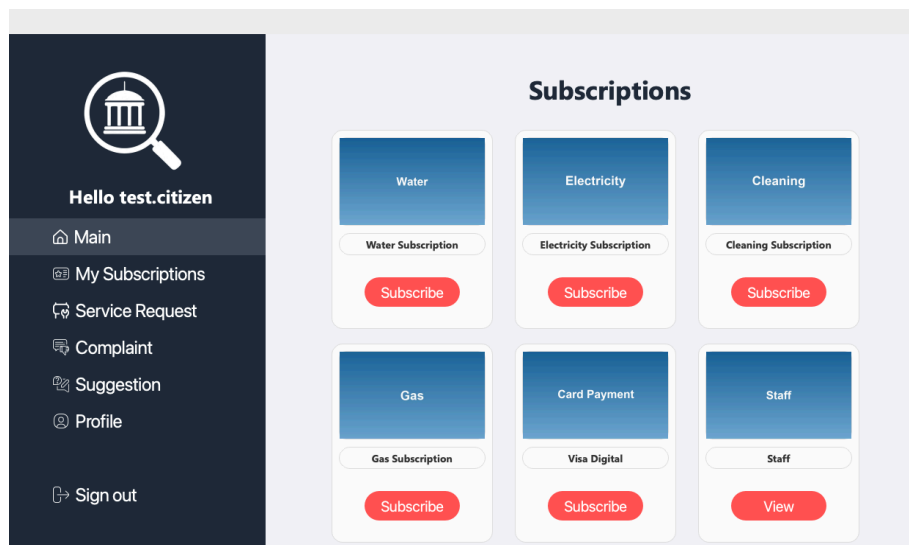
A **Username** field

A **Password** field (with hidden input)

A **Log in** button to authenticate credentials

If login is successful, the employee is redirected to the admin panel where they can manage users, subscriptions, and activities

-User Dashboard:



This is the main dashboard that appears after a user logs in. It provides access to all the core services via a sidebar and interactive subscription cards.

Key Features:

Left Navigation Panel:

Main: Displays available services

My Subscriptions: View active subscriptions

Service Request, Complaint, Suggestion: Submit forms

Profile: View or edit user info

Sign out: Return to the welcome screen

Main Area:

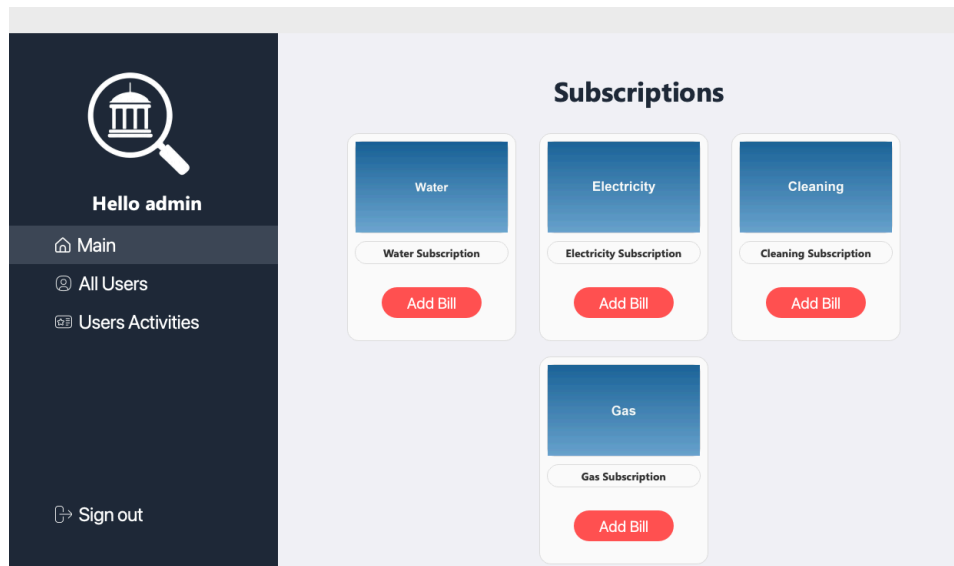
Displays six cards for various subscriptions:

Water, Electricity, Gas, Cleaning, Digital Visa, and Staff Info

Each card includes an image, label, and a **Subscribe** or **View** button

This interface is responsive and designed for easy navigation and fast access to services

-Employee Dashboard:



This interface is used by **municipal employees** (admins) to manage the system
The dashboard provides essential tools to handle subscriptions and monitor user activity

Key Features:

Left Sidebar Menu:

Main: Access bill creation forms

All Users: View and manage all registered users

Users Activities: Track user complaints, requests, and suggestions

Sign out: Exit to the welcome screen

Main Area (Subscriptions):

Shows cards for Water, Electricity, Cleaning, and Gas services

Each card includes an **Add Bill** button to create billing records for users

This dashboard simplifies employee tasks and ensures services are properly billed and tracked

Database Structure:

The application uses **SQLite** as its local database to store and manage all data. The structure is relational and follows normalization principles. Below are the core tables and their purposes

users: Stores registered users' basic information (username, name, surname, password)

employee: Contains login credentials for authorized staff members

bills: Stores billing data per service (e.g., water, electricity), linked to users by user_id

user_profiles: Saves extended user information such as contact details, gender, and city

requests: Contains service requests submitted by users, with status tracking

complaints: Stores user complaints including type, contact info, and description

suggestions: Allows users to suggest improvements and proposed solutions

Subscription-specific tables:

water_subscriptions

electricity_subscriptions

gas_subscriptions

cleaning_subscriptions

visa_subscriptions

Each of these stores form-specific data related to the corresponding service

-Conclusion:

The ***Municipal Services Tracking System*** stands as a comprehensive solution designed to enhance the interaction between citizens and municipal authorities through a unified, user-friendly platform. This project brought together critical functionalities such as user authentication, multi-service subscription management, bill generation, and real-time tracking of complaints, suggestions, and service requests

By implementing a clean graphical interface using **PyQt5** and leveraging the simplicity of **SQLite** for data management, the system offers both usability and efficiency. The separation between the user and employee interfaces ensures role-based access and control, improving security and administrative workflow

This project not only demonstrates strong technical implementation but also reflects a clear understanding of real-world municipal processes. It sets a solid foundation for future enhancements such as cloud integration, data analytics, mobile adaptation, and notification systems

Ultimately, this system bridges the gap between digital governance and everyday citizens, making municipal service management smarter, faster, and more accessible